

YUNE SANG LEE, Ph.D.

Research Professor · Seoul National University Brain Imaging Center
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Research Interests: Cognitive neuroscience of music and language; rhythm-based interventions for neurorehabilitation; digital sound therapy for cognitive and emotional health (binaural beats, immersive audio); aging brain and successful communication

EDUCATION & TRAINING

University of Pennsylvania, Philadelphia, PA 2011 – 2016
Department of Neurology & Center for Cognitive Neuroscience
Postdoctoral Research Fellow

Dartmouth College, Hanover, NH 2005 – 2010
Department of Psychological and Brain Sciences
Ph.D. in Cognitive Neuroscience

Yonsei University, Seoul, S. Korea 2001
Department of System Biology
B.S. in Biology

EMPLOYMENT

Seoul National University Oct. 2025 – Present
Research Professor
SNU Brain Imaging Center / Computational Clinical Science Lab

The University of Texas at Dallas, Richardson, TX Aug. 2020 – May 2025
Assistant Professor
Department of Speech, Language, and Hearing, School of Behavioral and Brain Sciences
Callier Clinical Research Center & Center for BrainHealth

The Ohio State University, Columbus, OH Aug. 2016 – July 2020
Assistant Professor
Department of Speech and Hearing Sciences / Chronic Brain Injury, Discovery Themes

T.K.B Commercial Music Studio, Seoul, S. Korea Mar. 2003 – July 2005
Music Director

Samsung Capital, Seoul, S. Korea Jan. 2002 – Feb. 2003
P.R. Officer

GRANTS

Extramural Funding — Awarded

Impact of Rigorous Music Training on Brain Development
Agency: Dallas Symphony Orchestra
Role: PI | Total: \$30,000 | Duration: 12/01/2023 – 11/30/2024

TheraBeat: A Novel Sound Therapy for Alzheimer's Patients
Agency: AWARE (awaredallas.org)
Role: PI | Total: \$40,000 | Duration: 05/01/2022 – 03/31/2024

SLAM Lab Research Support
Agency: Neuroscience Innovation Foundation (neurofdn.com)

Role: PI | **Total:** \$50,000 | **Duration:** 05/01/2022 – 04/30/2023

Non-invasive Brain Stimulation Using a Novel 3D Sound Therapy

Agency: Digisonic (digisonic.com)

Role: PI | **Total:** \$200,000 | **Duration:** 01/01/2022 – 12/31/2022

Speech Hero: A Rhythm-Based Speech Therapy App for Individuals with Aphasia

Agency: NIH Small Business Innovation Research (Grant No: 614362)

Role: Consortium PI | **Total:** \$50,000 | **Duration:** 06/01/2020 – 05/31/2021

Investigating the Neural Mechanisms of Language Recovery Through Rhythm Gaming Therapy in Aphasia

Agency: National Institutes of Health — NIDCD (Grant No: 1R21DC018699-01)

Role: PI | **Total:** \$429,000 | **Duration:** 01/01/2020 – 12/31/2022 * *No ESI advantage applied*

TRIPODS + X Project: An MBI TGDA + Neuro Program for Undergraduates

Agency: National Science Foundation (Grant No: NSF-CCF 1839356)

Role: co-PI (PI: Janet Best) | **Total:** \$200,000 | **Duration:** 09/01/2018 – 08/31/2023 ★ *Featured on NSF website front page*

Drum Dance Rehabilitation: A Novel Parkinson's Disease Therapy Program

Agency: The Parkinson's Foundation (Grant No: PF-CGP-19163)

Role: PI | **Total:** \$37,000 | **Duration:** 09/01/2019 – 08/31/2020

Application of fNIRS to Neuroscience Research of Speech, Language, and Music

Agency: Shimadzu Scientific Instrument, Japan

Role: PI | **Total:** \$380,000 | **Duration:** 12/18/2017 – 12/17/2018

Brain-based Evaluation of SoundMind — a Computerized Cognitive Training Program

Agency: Tech Incubator Program for Startup (TIPS) in Korea (Grant No: S2640193)

Role: PI (sub-contract) | **Total:** \$63,000 | **Duration:** 01/01/2019 – 12/31/2019

Music and Language Skills in Korean Children

Agency: Korea Advanced Institute of Science and Technology

Role: co-I (PI: Kyung-Myun Lee) | **Total:** \$20,000 | **Duration:** 09/01/2018 – 08/31/2020

Intramural Funding — Awarded

Investigating the Therapeutic Mechanisms of Binaural Beat Stimulation in Children with Developmental Language Disorder

Agency: School of Behavioral and Brain Sciences, University of Texas at Dallas

Role: PI | **Total:** \$12,500 | **Duration:** 01/01/2022 – 06/30/2022

Mobile EEG System Request

Agency: School of Behavioral and Brain Sciences, University of Texas at Dallas

Role: co-PI (PI: Maguire) | **Total:** \$20,785 | **Duration:** —

Investigating Compensatory Language Processes Prompted by Rhythm Video Game Therapy

Agency: Chronic Brain Injury Discovery Theme, OSU

Role: PI | **Total:** \$50,000 | **Duration:** 01/01/2017 – 12/31/2017

A Novel Parkinson's Disease Therapy Program for the Columbus Community

Agency: Connect & Collaborate Grant Program

Role: PI | **Total:** \$29,000 | **Duration:** 09/01/2019 – 08/31/2020

OSU Pilot Neuroimaging Grant

Agency: College of Arts and Sciences, OSU

Role: PI | **Total:** \$25,000 | **Duration:** 06/01/2018 – 05/31/2019

OSU Social and Behavioral Sciences Small Grant Program

Agency: College of Arts and Sciences, OSU

Role: PI | **Total:** \$4,000 | **Duration:** 09/01/2017 – 08/31/2018

OSU Social and Behavioral Sciences Special Grant

Agency: College of Arts and Sciences, OSU

Role: PI | **Total:** \$4,000 | **Duration:** 09/22/2018 – 09/21/2019

Travel Awards

ASHA Lesson for Success

Agency: American Speech-Language-Hearing Association

Role: Early-career investigator | **Total:** Full travel + registration | **Duration:** 04/27/2020 – 04/29/2020

The Humanities and the Arts Travel Grant Program

Agency: Global Arts and Humanities, Discovery Themes, OSU

Role: co-I (PI: David Huron) | **Total:** \$10,000 | **Duration:** 08/01/2017 – 09/01/2017

Consulting Service

Powerless, Wireless, and Nontoxic Bio-telemetry Sensor and Communication System

Agency: Korea Polytechnic University

Role: Consultant (PI: Jae-Young Chung) | Total: — | Duration: 02/01/2018 – 01/31/2021

PUBLICATIONS

Selected Publications

- Kim, J.H., Kim, H-W., Kovar, J., Lee, Y.S. (2024). Neural consequences of binaural beat stimulation on auditory sentence comprehension: an EEG study. *Cerebral Cortex*. <https://doi.org/10.1093/cercor/bhad459>
- Kim, H-W., Lee, K., Lee, Y.S. (2023). Sensorimotor and working memory systems jointly support development of perceptual rhythm processing. *Developmental Science*, 26(1): e13261.
- Heard, M.J. & Lee, Y.S. (2020). Shared neural resources of rhythm and grammar: An ALE Meta-Analysis. *Neuropsychologia*, 137: 107284.
- Lee, Y.S., Ahn, S.H., Holt, R.F., Schellenberg, G. (2020). Rhythm and syntax processing in school-age children. *Developmental Psychology*, 56(9): 1632–1641.
- Lee, Y.S., Turkeltaub, P.E., Granger, R.H., Raizada, R.D.S. (2012). Categorical speech processing in Broca's area. *Journal of Neuroscience*, 32(11): 3942–3948.
- Lee, Y.S., Janata, P., Frost, C., Hanke, M., Granger, R.H. (2011). Investigation of melodic contour processing in the brain using pattern-based fMRI. *NeuroImage*, 57(1): 293–300.

Peer-Reviewed Articles

- Henry, M.J., Obleser, J., Crusey, M.R., Fuller, E.R., Lee, Y.S., Meyer, M., et al. (2025). How strong is the rhythm of perception? A registered replication of Hickok et al. (2015). *Royal Society Open Science*, 12(6), 220497.
- Thibodeau, L., Freeman, E., Kronenberger, K., Suarez, E., Kim, H-W., Qi, S., Lee, Y.S. (2025). Exploration of auditory processing beyond speech recognition. *Audiology Research*, 15(3), 59.
- Rashidi, F., Parsaei, M., Kiani, I., Sadri, A., Aarabi, S.R., Lee, Y.S., Moghaddam, H.S. (2024). White matter correlates of impulsive behavior in healthy individuals: A diffusion MRI study. *Psychiatry and Clinical Neurosciences Reports*, 3(4), e70018.
- Schneider, J.M., Kim, J., Poudel, S., Lee, Y.S., Maguire, M. (2024). Environmental experiences and cognitive outcomes are predicted by resting-state EEG in school-aged children. *Developmental Cognitive Neuroscience*.
- Kim, H-W., Park, M.S., Lee, Y.S., Kim, C-Y. (2024). Prior conscious experience modulates the impact of audiovisual temporal correspondence on visual processing outside of awareness. *Consciousness and Cognition*.
- Kim, H-W., Kovar, J., Bajwa, J.S., Miran, Y., Ahmad, A., Moreno, M.M., Price, T.J., Lee, Y.S. (2024). Rhythmic motor behavior explains individual differences in grammar skills in adults. *Scientific Reports*. <https://doi.org/10.1038/s41598-024-53382-9>
- Kim, J.H., Kim, H-W., Kovar, J., Lee, Y.S. (2024). Neural consequences of binaural beat stimulation on auditory sentence comprehension: an EEG study. *Cerebral Cortex*. <https://doi.org/10.1093/cercor/bhad459>
- Kim, H-W., Kim, M., McLaren, K., Lee, Y.S. (2024). No influence of regular rhythmic priming on grammaticality judgment and sentence comprehension in English-speaking children. *Journal of Experimental Child Psychology*, 237, 105760.
- Kim, H-W., Happe, J., Lee, Y.S. (2023). Beta and gamma binaural beats enhance auditory sentence comprehension. *Psychological Research*. <https://doi.org/10.1007/s00426-023-01808-w>
- Kim, H-W., Lee, K., Lee, Y.S. (2023). Sensorimotor and working memory systems jointly support development of perceptual rhythm processing. *Developmental Science*, 26(1): e13261.
- Lee, Y.S., Rogers, C., Min, N.E., Wingfield, A., Grossman, M., Peelle, J.E. (2022). Neural compensation supporting successful speech comprehension in normal aging. *Aging Brain*, 2: 100051.
- Yoon, Y-S., Gutierrez, M., Lee, Y.S. (2021). Effects of the configuration of hearing loss on consonant perception. *Journal of American Academy of Audiology*, 32(08): 521–527.
- Heard, M.J., Li, X., Lee, Y.S. (2021). Hybrid auditory fMRI imaging. *Journal of Neuroscience Methods*, 358: 109198.
- Moritz, M., Heard, M., Kim, H-W., Lee, Y.S. (2020). Invariance of edit-distance to tempo in rhythm similarity. *Psychology of Music*.
- Lee, Y.S., Ahn, S.H., Holt, R.F., Schellenberg, G. (2020). Rhythm and syntax processing in school-age children. *Developmental Psychology*, 56(9): 1632–1641.
- Heard, M.J. & Lee, Y.S. (2020). Shared neural resources of rhythm and grammar: An ALE Meta-Analysis. *Neuropsychologia*, 137: 107284.
- Lee, K.M. & Lee, Y.S. (2019). Beat synchronization development of Korean elementary school students. *The Korean Society for Music Theory*, 26(2): 251–270.
- Lee, Y.S., Wingfield, A., Min, N.E., Kotloff, E., Grossman, M., Peelle, J.E. (2018). Differences in hearing acuity among 'normal-hearing' young adults modulate the neural basis for speech comprehension. *eNeuro*, 0263-17.2018. ★ Featured in US News & World Report
- Lee, Y.S. (2018). Impact of subtle hearing loss on the cognition of young adults. *Hearing Journal*, 71(10):30. [invited]

- Belyk, M., Lee, Y.S., Brown, S. (2018). How does human motor cortex regulate vocal pitch in singers? *Royal Society Open Science*, 5(8): 172208.
- Quandt, L.C., Lee, Y.S., Chatterjee, A. (2017). Neural bases of action abstraction. *Biological Psychology*, 129: 314–323.
- Lee, Y.S., Zreik, J., Hamilton, R.H. (2017). Patterns of neural activity predict picture-naming performance of a patient with chronic aphasia. *Neuropsychologia*, 94: 52–60.
- Lee, Y.S., Min, N.E., Wingfield, A., Grossman, M., Peelle, J.E. (2016). Acoustic richness modulates the neural networks supporting auditory sentence comprehension. *Hearing Research*, 333: 108–117.
- Lee, Y.S., Peelle, J.E., Kraemer, D., Lloyd, S., Granger, R.H. (2015). Multivariate sensitivity to voice during auditory categorization. *Journal of Neurophysiology*, 114(3): 1819–1826.
- Lee, Y.S., Janata, P., Frost, C., Martinez, Z., Granger, R.H. (2015). Melody recognition revisited. *Psychonomic Bulletin & Review*, 22(1): 163–9.
- Wiener, M., Lee, Y.S., Lohoff, F., Coslett, H.B. (2014). Individual differences in morphometry and activation of time perception networks are influenced by genotype. *NeuroImage*, 89: 10–22.
- Raizada, R.D.S. & Lee, Y.S. (2013). Smoothness without smoothing: why Naïve Bayes is not naïve for multisubject searchlight studies. *PLOS ONE*, 8(7): e69566.
- Lee, Y.S., Turkeltaub, P.E., Granger, R.H., Raizada, R.D.S. (2012). Categorical speech processing in Broca's area. *Journal of Neuroscience*, 32(11): 3942–3948.
- Lee, Y.S., Janata, P., Frost, C., Hanke, M., Granger, R.H. (2011). Investigation of melodic contour processing in the brain using pattern-based fMRI. *NeuroImage*, 57(1): 293–300.
- Russ, B.E., Lee, Y.S., Cohen, Y.E. (2007). Neural and behavioral correlates of auditory categorization. *Hearing Research*, 229: 204–12.

Manuscripts Under Review

- Kim, H-W., Kovar, J., Lee, Y.S. From rhythm to language: beta oscillations during tapping predict sentence comprehension.
- Wang, L-S, Wang, J-H, Chang, L-T, Lee, Y.S., Kung, C-C. Neural similarity and modularity are not necessarily oppsing properties.

Book Chapters

- Lee, Y.S., Wilson, M., Howland, K.M. (2024). Ch 6: Music for speech disorders. In *Music Therapy & Music-Based Interventions in Neurology*. Springer Nature.
- Lee, Y.S., Thaut, C., Santoni, C. (2019). Neurological music therapy for speech and language rehabilitation. *Oxford Handbook on Music and the Brain*. Oxford University Press.

CONFERENCE PRESENTATIONS (2016 – PRESENT)

29th Annual Meeting of the Korean Society for Brain and Neural Sciences (upcoming) <i>From rhythm to language: beta oscillations during tapping predict sentence comprehension</i> Kim, H-W, Kovar, J., Lee, Y.S. [Poster]	Sept. 2026
7th Neuroscience Forum for Alzheimer's Disease (NFAD) <i>New frontiers of digital sound therapy for neurorehabilitation</i>	Feb. 2024
Society for Neuroscience (SFN 2023) <i>The effect of beta binaural beat stimulation on resting-state brain activity in patients with MCI or AD</i> Leung, C., Kim, J.H., Happe, J., Kovar, J., Lee, Y.S. [Poster]	Nov. 2023
Society for Neuroscience (SFN 2023) <i>Role of the pre-supplementary motor area in syntax: a tACS study</i> Leung, C., Heard, M., McLaren, K., Wood, K., Lee, Y.S. [Poster]	Nov. 2023
Cognitive Neuroscience Society (CNS 2023) <i>Neural correlates of enhanced auditory sentence comprehension by binaural beat stimulation</i> Kim, J.H., Kim, H-W., Kovar, J., Lee, Y.S. [Poster]	Mar. 2023
Society for Neurobiology of Language (SNL 2022) <i>The influence of dopamine genotypes on the relationship between rhythm and language processing</i> Kim, H-W., Kovar, J., Bajwa, J., Mian, Y., Ahmad, A., Moreno, M.M., Price, T.J., Lee, Y.S. [Poster]	Oct. 2022
Society for Neurobiology of Language (SNL 2022) <i>The effects of rhythm priming on syntactic language processing in children</i> Kim, H-W., Ginter, K., Lee, Y.S. [Poster]	Oct. 2022
American Congress of Rehabilitation Medicine (ACRM 2022) <i>A novel sound therapy combining music and binaural beats for Alzheimer's disease</i> Lee, Y.S. [Symposium Talk]	Nov. 2022
Society for Music Perception & Cognition (SMPC 2022) <i>Sensorimotor synchronization and auditory working memory jointly predict rhythm perception in children</i> Kim, H-W., Lee, K.M., Lee, Y.S. [Podium Talk]	Aug. 2022

Society for Music Perception & Cognition (SMPC 2022) <i>Prior listening to binaural beats facilitates grammar processing during auditory speech tasks</i> Happe, J., Kim, H-W., Lee, Y.S. [Poster]	Aug. 2022
American Congress of Rehabilitation Medicine (ACRM 2019) <i>Home-based rhythm video gaming therapy for aphasia</i> Lee, Y.S. [Symposium Talk]	Nov. 2019
Society for Music Perception and Cognition (SMPC 2019) <i>Influence of rhythm and beat priming on receptive grammar task</i> Yen, S., Bendoly, D., Heard, H., Lee, Y.S. [Poster]	Aug. 2019
Organization of Human Brain Mapping Conference 2019 <i>Exploring a hybrid auditory fMRI protocol combining ISSS and multiband sequences</i> Heard, M., Li, X., Wolfe, T., Lee, Y.S. [Poster]	June 2019
International Conference of Music Perception and Cognition (ICMPC), Montreal <i>Spontaneous rhythm tapping predicts ability of working memory in children</i> Lee, Y.S., Corbeil, K., Ahn, S.H. [Symposium Talk]	July 2018
International Conference of Music Perception and Cognition (ICMPC), Montreal <i>Shared neural resources of rhythm and grammar: An ALE meta-analysis</i> Heard, M.J. & Lee, Y.S. [Symposium Talk]	July 2018
Society for Neurobiology of Language, Baltimore, MD <i>Differences in hearing acuity among young adults modulate the neural basis for speech communication</i> Lee, Y.S., Wingfield, A., Min, N.E., Jester, C., Kotloff, E., Grossman, M., Peelle, J.E. [Poster]	Nov. 2017
Society for Neurobiology of Language, Baltimore, MD <i>Rhythm sensitivity assists in overcoming acoustic and syntactic challenges during speech listening</i> Ahn, S., Golthwaite, I., Corbeil, K., Byrer, A., Perry, K., Polakampalli, A., Miller, K., Holt, R.F., Lee, Y.S. [Poster]	Nov. 2017

INVITED TALKS

2026

The Society for Clinical Music Science Research (upcoming) <i>When music Therapy Meets Neuroscience: New Perspectives on Neurorehabilitation</i>	Oct. 2026
Colloquium, School of Digital Humanities and Computational Social Sciences, KAIST (upcoming) <i>Beyond background music: Functional Sound in the Age of Digital Technology</i>	Oct. 2026
Colloquium, Department of Music Therapy, Hansei University (upcoming) <i>Music Therapy Meets Neuroscience: New Frontiers in Neurorehabilitation</i>	Aug 2026
Korean Music Therapy Association (upcoming) <i>Music Therapy Meets Neuroscience: New Frontiers in Neurorehabilitation</i>	Aug 2026
Music, Mathematics, & Language workshop <i>Shared Circuits: How music and language meet in the brain</i>	June 2026
23rd Avison Biomedical Symposium 2026 <i>Functional Sound as Neuromodulation: Turning Brain Rhythms into Mental Regulation</i>	June 2026
Korean Brain Research Institute (KBRI) <i>From Rhythm to Brain Plasticity: Harnessing Music to Transform the Brain</i>	May 2026
Colloquium, Department of Brain & Cognitive Sciences, KAIST <i>Listening to the brain: towards a neuroscience of music for precision mental health</i>	Apr. 2026

2025

Seminar, College of Electronics and Information Engineering, Kwangwoon University <i>New frontiers of digital medicine: the power of functional sound</i>	Dec. 2025
Seminar, Department of Communication Disorders, Ewha Woman's University <i>Shared circuits: how music and language meet in the brain</i>	Oct. 2025
Colloquium, Department of Music Therapy, Ewha Woman's University <i>Beyond metronome: rhythm and the art of intervention</i>	Sept. 2025
Online Lecture for the Inner Communication and Meditation <i>The Brain, Music and Well-being: BMW story</i>	Aug. 2025
Keynote, Society for Clinical Music Science Research <i>The Brain, Music and Well-being: BMW story</i>	July 2025
1st PKSA (Philadelphia Korean Scholar Association) Korea Workshop <i>The Brain, Music and Well-being</i>	Jan. 2025

2024

Neuroscience Forum on Alzheimer's Disease, S. Korea <i>New frontiers of digital sound therapy for neurorehabilitation</i>	Feb. 2024
Science Concert, S. Korea <i>Positive impact of music on brain health</i>	Feb. 2024
Dallas Symphony Orchestra Guild, Dallas, TX <i>How does musical training impact brain development of school-age children?</i>	Jan. 2024
2023	
CHI Memorial Hospital, Chattanooga, TN <i>Non-invasive brain stimulation using binaural beat combined with music</i>	Sept. 2023
Vanderbilt University Medical Center, Nashville, TN <i>The Brain, Music, and Wellbeing</i>	Sept. 2023
Neuroscience Innovation Foundation, St. Louis, MO <i>A novel sound therapy for Alzheimer's disease</i>	Apr. 2023
The Dallas Symphony Orchestra, Dallas, TX <i>The brain, music, and well-being</i>	Feb. 2023
2022	
Invited Seminar, Seoul National University Hospital, Bundang <i>The brain, music, and well-being</i>	Nov. 2022
Keynote, AWARE Membership Kick-off Event <i>A novel sound therapy for Alzheimer's disease</i>	Aug. 2022
Invited Seminar, Laboratory for Cognition and Neurostimulation, University of Pennsylvania <i>The brain, music, and well-being</i>	June 2022
Special Lecture, Department of Brain & Cognitive Sciences, Seoul National University <i>The brain, music, and well-being</i>	May 2022
Dartmouth Symposium on Music and Medicine in Cross-Cultural Perspective <i>The brain, music, and well-being</i>	Apr. 2022
Frontiers Talk Series, Center for BrainHealth <i>The brain, music, and well-being</i>	Mar. 2022
2021	
Center for Vital Longevity Virtual Science Series <i>The brain, music, and well-being</i>	Apr. 2021
NIH Music & Health Investigator Meeting <i>Investigating the neural mechanisms underlying language recovery through rhythm therapy in aphasia</i>	Mar. 2021
2020	
Therapy Insights (Continuing Education Course) <i>Beat the beat: Rhythm-based intervention for communication disorders</i>	Dec. 2020
Colloquium, Music and Health Science Research Collaboratory, University of Toronto <i>New frontiers in neurologic music therapy</i>	Feb. 2020
Colloquium, Cognitive and Behavioral Science, Neuroscience and Music Dept., Washington and Lee University <i>The brain, music, and well-being</i>	Feb. 2020
2019	
Otolaryngology Department Seminar, Washington University in St. Louis, MO <i>The neural and genetic mechanisms underlying the connection between music and language</i>	May 2019
Topology, Geography, and Data Seminar, OSU <i>The connection between speech, language, and music</i>	Jan. 2019
2018	
Philadelphia Korean Scholar Association (PKSA) Symposium [Keynote] <i>The connection between speech, language, and music</i>	Dec. 2018
Korea Society for Music Perception and Cognition (KSMPC) [Keynote] <i>The connection between speech, language, and music</i>	May 2018
Annual Brain Health & Performance Summit, Columbus, OH <i>Rhythm-based rehabilitation for communication disorders</i>	Mar. 2018
Annual TBI Summit, Columbus, OH <i>The brain, music, and well-being</i>	Mar. 2018
Cognitive Proseminar, Dept. of Psychology, OSU <i>The connection between speech, language, and music</i>	Jan. 2018

2017

KSEA SW Ohio Chapter Seminar, University of Cincinnati <i>The brain, music, and well-being</i>	Dec. 2017
"Aging in Community" International Standardization Symposium, S. Korea <i>Neural compensation supporting successful communication in normal aging</i>	Dec. 2017
KSEA Ohio Chapter Seminar, The Ohio State University <i>Mind decoding: New frontiers in neuroimaging</i>	Apr. 2017

2016

Eye and Ear Institute, The Ohio State University <i>Predicting CI outcome using novel neuroimaging techniques</i>	Dec. 2016
OSU Music Cognition Group, The Ohio State University <i>Rhythm-based rehabilitation for communication disorders</i>	Sep. 2016
The Haskins Laboratories, New Haven, CT <i>New frontiers in brain-based rehabilitation for communication disorders</i>	May 2016

AD HOC REVIEWER

American Journal of Speech-Language Pathology · Brain & Language · Cerebral Cortex · Clinical Archives of Communication Disorders · Cortex · European Journal of Neuroscience · Experimental Brain Research · Frontiers in Human Neuroscience · Hearing Research · Human Brain Mapping · International Journal of Psychophysiology · Journal of Alzheimer's Disease · Journal of Cognitive Neuroscience · Journal of Cognitive Science · Journal of Neuroscience · Journal of Neurophysiology · Journal of Neurology · Magnetic Resonance Imaging · Music Perception · NeuroImage · NeuroImage Clinical · Neurocase · Psychomusicology · Scientific Reports

COURSES TAUGHT

Neural Basis of Speech, Language, and Music (undergraduate seminar)	Spring 2021 – 2025
Neural Basis of Speech, Language, and Music (graduate seminar)	Fall 2020 – 2025
Communication, Cognition, and the Aging Brain (undergraduate seminar)	Fall 2023 – 2025
Neurology of Speech and Hearing — SHS 5760, OSU	Fall 2017 – Fall 2019
NeuroImaging for Speech, Language, and Music — SHS 8950, OSU	Spring 2017 – 2019
Statistics — SHS 6786, OSU	Spring 2017 – 2019

Trainees

Matthew Heard, doctoral student	2018 – 2023
Hyun-Woong Kim, doctoral student	2019 – 2024
Carole Leung, doctoral student	2022 – 2024
Saranya Mundayoor, doctoral student	2023 – 2024
Jea-Hong Kim, postdoc	2021 – 2023

UNIVERSITY / COMMUNITY / ACADEMIC SERVICE (2016 – PRESENT)

Director of Scientific Affairs, The Society for Clinical Music Science Research	Jan. 2026 – present
Director of Scientific Affairs, Korean Society for Music Perception & Cognition	Sep. 2016 – present
Callier Postdoc Hiring Committee	Jan. 2024
BBS Committee for Cluster Hire in Lifespan Initiative	Sep. 2022 – 2025
Panelist, National Youth Leadership Forum: Explore STEM Program	June 2022
UTD Intellectual Property Committee	Sep. 2021 – 2025
SLHS Ph.D. Program Admission Committee	Nov. 2021 – 2025
ASHA 2021 Convention Topic Committee	Feb. 2021 – Nov. 2021
BBS Website Renovation Committee	Apr. 2021 – Dec. 2022
BBS Delegate for Blackstone Launchpad Stewardship Council	Aug. 2020 – Aug. 2025
Doctoral Dissertation Committee for Christina Dugas	Aug. 2020 – Aug. 2023
Doctoral Dissertation Committee for Kathryn Kriedler	Dec. 2020 – Feb. 2024
External Reviewer, Swiss National Scientific Foundation	May 2021
External Reviewer, Clinical and Translational Science Awards, Auburn University	Feb. 2020
Session Chair, ACRM Mini-Symposium	Nov. 2019
President, KSEA Ohio Chapter	July 2019 – July 2020
External Reviewer, Clinical and Translational Science Awards, University of Washington	Oct. 2018
Faculty Mentor, OSU STEP (Second-year Transformational Education Program)	Sep. 2017 – Apr. 2019

CBI-Sponsored Monthly fNIRS Seminar Series
Session Chair, ICMPC
Reviewer for STEAM Powered Project Grants

Sep. 2017 – 2025
July 2018
July 2017

MEDIA COVERAGE /Interview

쇼츠 릴스에 빼앗긴 문해력 MBC 뉴스데스크	Mar. 2026
음악을 듣고 연주할 때 뇌에서는 어떤 일이 일어나는가? RYM Collabs.	Mar. 2026
Binaural Beats Boost Language Skills — Neuroscience News	Dec. 2023
Neural Nourishment: How Discovering New Music May Garner Some Serious Brain Regions — Vox	Sep. 2022
Music: A Brain Enhancer? — TwoMaverix	Mar. 2022
NSF: Featuring our fMRI Study Investigating Rhythm and Grammar — NSF Website	Sep. 2018
Podcast Interview with Parsing Neuroscience	Aug. 2018
How Subtle Hearing Loss While Young Changes Brain Function — US News	June 2018
A Minor Hearing Loss Now Can Mean Major Dementia Risk Later — Healthline	July 2018
Early Hearing Loss Could Pave the Way for Dementia — Medical News Today	May 2018
Subtle Hearing Loss While Young Alters Brain Function — Neuroscience News	May 2018